

Feature Interactions Under Distribution Shift

An Exploratory Study of Simultaneous Feature Shifts and Model Performance

Research report

Dataset: UCI Bike Sharing Dataset (hourly data)

Primary models: Linear Regression and Random Forest Regressor

1. Abstract

This study investigates whether simultaneous distribution shifts in multiple input features affect model performance in a way that can be explained simply by adding their individual effects. Using the hourly UCI Bike Sharing Dataset, we trained regression models to predict total bike rentals and evaluated them on an unchanged test distribution and on test sets in which selected environmental features were shifted. For each feature pair, individual and combined shifts were compared using mean absolute error (MAE). The main Linear Regression experiment found a negative interaction for temperature and humidity, while temperature-windspeed and humidity-windspeed interactions were close to zero. The temperature-humidity pattern remained negative across different train/test splits, with a Random Forest model, and under an alternative boundary-safe shift definition. However, the magnitude changed with the shift definition. The results therefore provide an exploratory finding within this dataset and experimental setup rather than a universal claim about feature interactions.

2. Introduction

Machine-learning models are commonly evaluated under the assumption that future data will resemble the data used during development. In practice, the distribution of input features can change after deployment. This phenomenon is commonly described as distribution shift. When more than one feature changes simultaneously, the effect on model performance may not be equal to the simple sum of the effects produced by each feature separately.

This project studies that question experimentally. Rather than asking only whether a model becomes less accurate after a shift, the study asks whether the combined effect of two shifting features is additive or whether their simultaneous change produces a different outcome.

3. Research Question

When multiple features shift simultaneously, is the resulting model-performance degradation simply the sum of their individual effects, or do interactions between shifting features change the outcome?

The study focuses on three environmental feature pairs: temperature and humidity, temperature and windspeed, and humidity and windspeed.

4. Dataset

The study uses the UCI Bike Sharing Dataset, specifically the hourly dataset *hour.csv*. It contains 17,379 observations and 17 columns and has no missing values in the analysed data.

The target variable is **cnt**, the total number of rented bikes. The model features used were temperature (**temp**), humidity (**hum**), windspeed (**windspeed**), hour (**hr**), season, workingday, and weather situation (**weathersit**). The variables **casual** and **registered** were excluded because they are components of the target variable and would create target leakage. The record identifier and raw date were also excluded from the baseline feature set.

5. Methodology

The workflow consisted of data exploration, feature selection, train/test splitting, baseline model training, controlled test-distribution shifts, performance evaluation, interaction calculation, and robustness checks.

Baseline: A Linear Regression model was trained on 80% of the data and evaluated on the remaining 20%, using a fixed random seed of 42 for the primary experiment.

Distribution shift: For the primary experiment, a selected test-set feature was shifted upward by 0.10 and clipped to the valid $[0, 1]$ range. The trained model was not retrained after the shift. This simulates a model encountering changed future input data.

Interaction calculation: If E_A is the individual effect of shifting feature A and E_B is the individual effect of shifting feature B, the expected additive effect is $E_A + E_B$. The interaction is defined as the actual combined effect minus this expected additive effect. A value near zero indicates approximately additive behaviour; a negative value indicates a combined effect smaller than the additive expectation; a positive value indicates a larger combined effect.

6. Baseline Performance

Model	Baseline MAE	Baseline R ²
Linear Regression	106.961	0.343
Random Forest	46.901	—

The lower Random Forest MAE shows that the second model fits this prediction task better in absolute error. However, baseline accuracy is not the central research quantity; the study is primarily concerned with how performance changes under controlled shifts.

7. Main Experimental Results

Feature pair	Interaction at +0.10	Interpretation
Temperature + Humidity	−3.782	Negative / non-additive
Temperature + Windspeed	+0.189	Approximately additive
Humidity + Windspeed	−0.179	Approximately additive

For temperature and humidity, the baseline MAE was 106.961. A temperature-only shift increased MAE to 116.499, corresponding to an effect of +9.538. A humidity-only shift reduced MAE to 103.181, corresponding to an effect of −3.780. The additive expectation was therefore +5.758, but the combined shift produced MAE of 108.937, an actual effect of +1.976. The resulting interaction was −3.782.

8. Robustness and Sensitivity Checks

Different train/test splits: For Linear Regression with the +0.10 shift, the temperature-humidity interaction was -3.782 for seed 42, -3.870 for seed 21, and -4.051 for seed 7. The direction remained negative across all three tested splits.

Second model: A Random Forest Regressor was also tested. For the temperature-humidity pair at +0.10, its interaction was -1.819 . The magnitude differed from Linear Regression, but the direction remained negative.

Boundary-safe shift: Because additive +0.10 shifts can be clipped at the upper boundary, an alternative shift was tested using $x + 0.10(1 - x)$. This moves each value 10% of the remaining distance toward 1 without clipping. Under this alternative, the temperature-humidity interaction was -0.828 . The negative direction therefore persisted, although the magnitude changed.

9. Additional Analysis and Interpretation

The temperature-humidity correlation in the dataset is weak (approximately -0.070), so the observed result is not easily explained simply by strong correlation between the two shifted features.

For the Linear Regression model, the average prediction changes produced by the two individual shifts were additive: temperature shifted predictions by approximately $+28.831$ on average, while humidity shifted them by approximately -21.514 . The combined average change was approximately $+7.317$, equal to the sum of the individual average changes. However, MAE is based on absolute row-level errors, so additive prediction changes do not necessarily produce additive changes in MAE. This provides a plausible mechanism for the observed MAE interaction, but it is not claimed as a complete proof of the underlying mechanism.

10. Discussion

The experiments suggest that the effect of simultaneous feature shifts depends on the feature pair. Temperature and humidity showed the clearest non-additive behaviour in the tested setup, whereas the other two environmental pairs produced interactions close to zero.

Importantly, the result should not be interpreted as proof that temperature and humidity have a universal interaction under distribution shift. The experiment measures the behaviour of specific trained models, on one dataset, under specific artificial shift definitions and magnitudes. The changing interaction magnitude under the two shift definitions also shows that the way distribution shift is constructed matters.

11. Limitations

- The study uses one dataset, so the findings cannot automatically be generalized to other domains.
- Only a small number of feature pairs were tested.
- The primary shift is an artificial +0.10 perturbation rather than an empirically observed deployment shift.
- Clipping at the upper boundary affects some observations, especially humidity; an alternative boundary-safe shift was therefore tested.
- Only a small set of model types and train/test seeds were evaluated.
- The interaction measure is based on changes in MAE. A non-zero MAE interaction does not by itself prove an explicit nonlinear feature interaction inside the model.

12. Conclusion

Within the tested UCI Bike Sharing setup, simultaneous shifts in temperature and humidity produced a consistently negative interaction in MAE. The temperature-windspeed and humidity-windspeed pairs were approximately additive. The temperature-humidity pattern persisted across multiple train/test splits, in a Random Forest robustness check, and under an alternative boundary-safe shift, although its magnitude changed. Overall, the study provides evidence that performance changes under simultaneous distribution shift cannot always be assumed to equal the sum of individual shift effects.

13. Future Work

- Test the methodology on additional datasets from different domains.
- Evaluate more feature pairs and larger sets of simultaneous shifts.
- Test multiple shift magnitudes and more realistic shifts derived from time periods or external conditions.
- Compare additional model families and performance metrics.
- Investigate whether the observed effects persist when model predictions and row-level error distributions are analysed with more formal statistical methods.

14. References

1. UCI Machine Learning Repository. Bike Sharing Dataset.
<https://archive.ics.uci.edu/dataset/275/bike%2Bsharing%2Bdataset>
2. scikit-learn documentation. LinearRegression, RandomForestRegressor, train_test_split, mean_absolute_error, and r2_score.
3. Fanaee-T, H. and Gama, J. (2014). Event labeling combining ensemble detectors and background knowledge. Progress in Artificial Intelligence, 2, 113–127.

Note: This report is an exploratory research study. Numerical results are reported from the finalized notebook experiments and rounded for presentation.